FE1101

Introduction to Food Business and Development

FULL MARKING SCHEME

Examination Session	Summer 2025
Paper Number	1
Duration	1.5 hours
Instructions to Candidates	Answer one question from each section. All questions carry equal marks.
Sections Covered	Section One — Co-operatives · Section Two — Development
Internal Examiners	Dr Bridget Carroll · Dr Stephen Onakuse · Dr Emma Beacom

Model answers, mark allocations and examiner notes for all four questions across both sections, so revision is complete regardless of which pair you are asked to answer.

SECTION ONE — CO-OPERATIVES

Question 1

50% of paper

QUESTION AS SET

Being member-based is a key strength that keeps co-operatives focused on serving their members. Discuss this statement.

VERDICT

Substantially agree — but with one important qualification. Member-ownership structurally aligns a co-operative's incentives with its members' needs; whether that alignment actually keeps the co-operative "focused" in practice depends on governance quality, scale, and how the focus on members is balanced against other principles.

FULL MODEL ANSWER

1. Ownership, control and benefit are unified in the same group

A co-operative is defined as a business that is **owned and democratically controlled by the people who use its services and share in its benefits** — it is a member-owned business, not an investor-owned one. This is the structural feature the statement is really pointing at: in a conventional company, the people who own the business (shareholders seeking a return on capital) are usually a different group from the people who use it (customers seeking a service). In a co-operative, ownership, control, and use collapse into a single group — the members. Because the owners *are* the users, there is no separate shareholder interest pulling the business away from what members actually need.

This is visible directly in the standard comparison of the two ownership models: in a user-owned business, ownership is held by the members, control is democratic, benefits are distributed equitably to members, and the reason for the service existing is to suit members' needs; in a conventional business, ownership sits with investors, control tracks shareholding, benefits flow to investors and private interests, and the reason for the service existing is primarily profit-making. The "member focus" the statement describes is not a stated value bolted onto the business — it falls directly out of who legally owns and controls it.

2. The co-operative principles operationalise this alignment

The seven co-operative principles — adopted by the International Co-operative Alliance in its 1995 Statement on the Co-operative Identity — exist specifically to keep this alignment intact as the business grows. Three do the heaviest lifting here:

- **Voluntary and open membership** ties eligibility to actual use of the service, not to capital or connections — anyone who can use the co-op's services and accept the responsibilities of membership may join.
- **Democratic member control** operates on a one-member-one-vote basis (not one-share-one-vote), exercised through the AGM and elected committees — so voting power tracks membership, not capital contributed.

- **Member economic participation** means members contribute capital but receive only limited compensation on it; surplus is instead distributed largely according to *use* of the co-operative (the patronage refund), not according to shareholding.

Together these three principles mean a co-operative cannot easily be captured by whoever contributes the most capital — voice and reward both track member *use*, which is exactly the mechanism that keeps a co-operative oriented toward serving its members rather than maximising return to outside investors.

3. The historical and economic rationale supports the statement directly

Co-operatives characteristically emerge at points in a supply chain where a group is poorly served by conventional business. Producers — farmers, fishers, craft workers — have historically formed co-operatives where they had little bargaining power relative to a small number of buyers or processors; by combining, they achieved better prices through collective input purchasing, processing and marketing. Consumers, particularly in Northern Europe and the UK, formed co-operatives where they felt disadvantaged relative to retailers, pooling their purchasing power to secure reliable, lower-cost food. In both cases, the co-operative model was chosen *because* it locks in a member-serving focus that a conventional, investor-owned business in the same market was not delivering.

> **Sourcing note.** *The ICA's 1995 Statement on the Co-operative Identity and the "food business chain" rationale (Box 2.4) are drawn from the module's own co-operatives reading; the one-member-one-vote and patronage-refund mechanics are directly from the Section 2 lecture material.*

4. The qualification: "focus" is a tendency the structure enables, not a guarantee

The statement should not be accepted unconditionally. The co-operative sector is, in the module's own words, a "broad church" — not every co-operative places the same weight on the member-empowerment ethos. Some co-operatives are formed primarily as a practical vehicle for job creation or income generation, where the values of solidarity and democracy are secondary to instrumental goals. The seventh principle, **Concern for Community**, explicitly directs surplus and effort beyond the immediate membership — toward sponsorship, environmental protection, and local job creation — which means an excellent co-operative is, by its own principles, not *purely* member-focused; it deliberately balances member interest against wider community interest. Scale matters too: as a co-operative grows, active member participation in AGMs and committees can weaken, creating a governance gap between the formal one-member-one-vote structure and members' actual influence over day-to-day decisions — the same kind of principal-agent distance that member-ownership is supposed to prevent.

Conclusion: the statement is well-founded as a description of what the co-operative *structure* is designed to do — unifying ownership, control and benefit in the member base is the specific mechanism that keeps a co-operative oriented toward its members rather than outside capital. But whether this structural design actually translates into sustained member focus in a given co-operative depends on governance quality, the co-operative's stated priorities within a diverse sector, and how it balances Principle 7's community obligations against Principles 1–3's member focus. The strength is real and structural; the guarantee is not automatic.

Mark Allocation

Band	Range	Criteria
First Class	70–100%	Clearly explains how member-ownership unifies ownership, control and benefit, AND connects this to Principles 1–3 (open membership, democratic control, patronage-based surplus) with correct mechanics (one-member-one-vote; patronage refund), AND grounds the argument in the historical/food-chain rationale for why co-operatives form, AND explicitly develops the qualification (governance quality, scale, Principle 7 balance) rather than treating the statement as unconditionally true.
Upper Second (2.1)	60–69%	Correctly explains the ownership/control/benefit alignment and at least two of Principles 1–3, with some qualification offered, but the qualification is asserted rather than developed with evidence.
Second Class / Pass	40–59%	Describes what a co-operative is and names some principles, but does not clearly connect member-ownership to the specific "focus on members" mechanism the question asks about; agrees with the statement with little critical examination.
Fail	0–39%	Generic description of co-operatives with no engagement with the specific statement; principles listed without explanation of how they relate to member focus; no reasoning about strengths or limits.

Question 1 total — 100% (50% of overall paper)

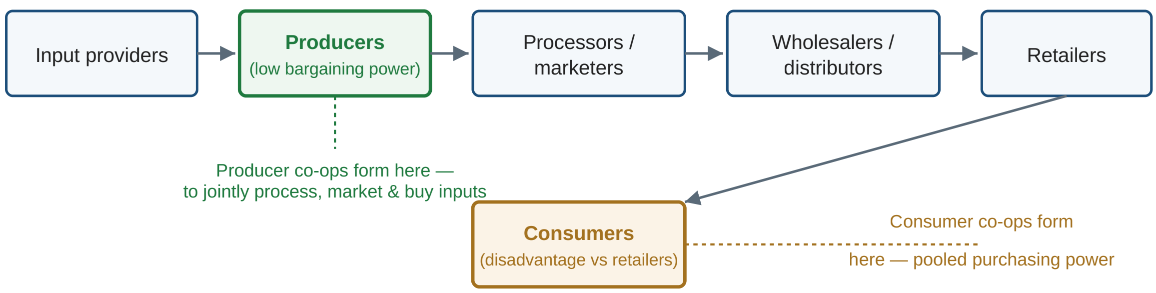
Question 2

50% of paper

QUESTION AS SET

Imagine you are working at a local agricultural co-operative (of any type). You have been asked to highlight the range of benefits of co-operative membership to the members. Identify and discuss five main benefits of the co-operative and justify your choices.

FULL MODEL ANSWER



Co-operatives emerge at the point in the food business chain where a group is poorly served by conventional business — most often producers with weak bargaining power, or consumers facing concentrated retail/distribution power.

Figure 1. Co-operatives characteristically form at the point in the food business chain where a group is poorly served by conventional business — this is the source of most of the benefits below.

The five benefits below are organised from the most direct economic benefit to the more structural/governance benefits, so that a moderator can see the range is genuinely distinct rather than five re-phrasings of "better prices."

Benefit 1 — Stronger bargaining power through collective marketing and processing

Individual small producers typically have little bargaining power relative to a comparatively small number of processors or buyers. By combining into a co-operative, members can process and market their output collectively, capturing a share of the value added further up the chain instead of accepting whatever price an individual buyer offers. *Justification:* this is the single most commonly cited historical reason producer co-operatives formed in the first place — it directly answers the "why join" question a real member would ask.

Benefit 2 — Lower input costs through collective purchasing

Members can jointly purchase inputs — feed, fertiliser, machinery, finance — at a better price than any one member could negotiate alone, since the co-operative is buying at volume on behalf of the whole membership. *Justification:* this benefit is realised before a member's product is even sold, so it improves margins regardless of output-market conditions, making it a distinct benefit from Benefit 1 rather than the same mechanism applied twice.

Benefit 3 — A direct, democratic voice in how the business is run

Because control operates on a one-member-one-vote basis exercised at the AGM and through committees, an ordinary member has a real say in the co-operative's direction regardless of how much capital they have contributed — unlike a conventional supplier relationship, where a small producer has no say at all in how the buyer runs its business. *Justification:* this is a governance benefit, not an economic one, and it matters because it means services can be shaped around what members actually need rather than what is most profitable for an external owner.

Benefit 4 — A more equitable, use-based share of the surplus

Members usually receive only limited compensation on the capital they have contributed; the co-operative's surplus is instead distributed largely according to each member's *use* of the co-operative (the patronage refund), rather than according to shareholding. *Justification:* this rewards members in proportion to how much they actually rely on the co-operative for their livelihood, which is a fundamentally different distribution logic from a conventional dividend and is worth identifying as its own benefit rather than folding into Benefit 3.

Benefit 5 — Access to services, security and community investment that would not otherwise exist

Especially in thin or remote markets, a co-operative can provide services — access to markets, credit, extension advice, or in community co-operative cases, essential local services — that would be uneconomic for a conventional business to offer. The Concern for Community principle also directs part of the co-operative's activity toward sponsorship, local job creation and environmental protection, which benefits members in their capacity as community residents as well as as producers. *Justification:* this captures the benefits that persist even where a member's individual trading volume is small, and explains why co-operatives are historically important in remote and economically marginal areas (e.g. the west of Ireland).

> **Sourcing note.** *All five benefits are grounded directly in the Section 2 lecture content on co-operative principles and the food business chain rationale (Box 2.4). No external claims beyond the provided course material were required for this answer.*

Mark Allocation

Band	Range	Criteria
First Class	70–100%	Identifies five genuinely distinct benefits (not overlapping restatements of the same mechanism), each with a clear explanation of the underlying mechanism AND an explicit justification of why it counts as a "main" benefit; correctly distinguishes economic benefits (bargaining power, input costs) from governance/structural ones (voice, patronage-based surplus, community access).
Upper Second (2.1)	60–69%	Five relevant benefits identified with reasonable discussion, but justification for one or two is thin, or two benefits overlap substantially.
Second Class / Pass	40–59%	Fewer than five distinct benefits identified, or benefits described in generic terms without being tied to the specific mechanics of co-operative membership (voting, patronage refund, collective purchasing).
Fail	0–39%	Little or no structured engagement with "benefits of co-operative membership"; answer describes co-operatives generally without identifying distinct, justified benefits.

Question 2 total — 100% (50% of overall paper)

SECTION TWO — DEVELOPMENT

Question 3

50% of paper

QUESTION AS SET

The Sustainable Development Goals aim to eradicate poverty by 2030, but growing inequality is one of the biggest social, economic and political challenges of our time. Discuss two elements underpinning the Sustainable Development Goals.

Examiner's note. *This question closely mirrors a Continuous Assessment prompt used earlier in the same module ("Discuss in relation to the elements underpinning the Sustainable Development Goals") — the lecture material (Dr Stephen Onakuse) names the three elements explicitly, so the framework below is not an outside interpretation but the module's own defined answer structure. The question asks for* **two** *of the three; the model answer below develops Universality and Integration, as these connect most directly to the inequality theme in the stem, and closes with a short note on why Transformation is an equally valid alternative choice.*

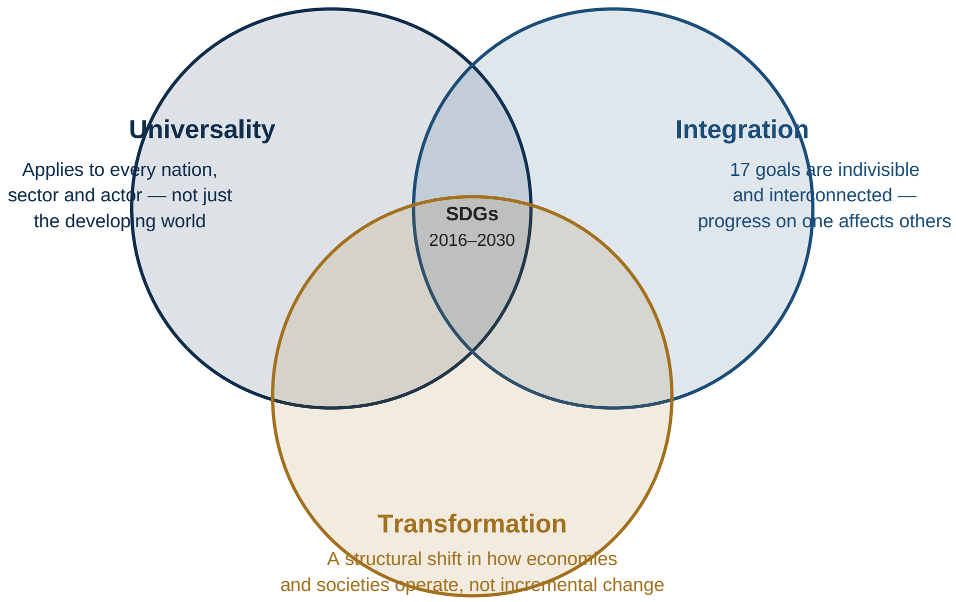


Figure 2. The three elements the module identifies as new and distinctive about the SDG framework, relative to the preceding Millennium Development Goals.

FULL MODEL ANSWER

The Sustainable Development Goals (SDGs) are a set of 17 goals for the world's future to 2030, backed by 169 detailed targets, negotiated over two years at the United Nations and agreed by nearly all the world's nations on 25 September 2015. What is new and different about the SDGs, relative to the Millennium Development Goals that preceded them, is captured in three elements: **Universality**, **Integration**, and **Transformation**. Two are discussed in depth below.

Element 1 — Universality

Unlike the Millennium Development Goals, which were framed largely as targets for developing countries to meet with donor support, the SDGs apply to *every* nation and every sector — cities, businesses, schools and organisations are all challenged to act, not just low-income states. This directly reframes the inequality problem posed in the question: growing inequality is not treated as a developing-world problem to be solved by aid, but as a shared global challenge. SDG 10 explicitly targets the reduction of inequality *within and among* countries — meaning inequality inside wealthy nations (income inequality, regional disparities) is as much a target of the framework as inequality between rich and poor countries. Universality is precisely what allows the SDGs to hold high-income countries accountable for the domestic inequality driving political instability there, rather than treating "poverty and inequality" as something that happens elsewhere.

Element 2 — Integration

The 17 goals are explicitly designed as indivisible and interconnected: progress (or failure) on one goal affects the others. This matters directly for the poverty/inequality tension in the question, because it means poverty (SDG 1) cannot be tackled in isolation from, for example, decent work and economic growth (SDG 8), reduced inequality (SDG 10), or climate action (SDG 13). A narrow anti-poverty programme that ignores structural inequality — in access to education, health, or economic opportunity — will struggle to deliver lasting poverty eradication, because the goals are built to reinforce or undermine one another. Integration is the element that explains *why* eradicating poverty by 2030 is harder than raising incomes alone: it requires simultaneous progress on the goals that determine whether income gains are equitably distributed and sustained.

> **Sourcing note.** *The 17 goals / 169 targets / 25 September 2015 adoption date, and the Universality · Integration · Transformation framing, are taken directly from the Dr Stephen Onakuse SDG lecture material provided for the module. SDG 10's "within and among countries" wording is the UN's own goal title, reproduced from the same slide set.*

A note on the third element (Transformation)

A candidate could equally have chosen Transformation as their second element: the SDGs are framed not as incremental improvement but as a structural shift in how economies and societies operate — which matters for inequality because closing an inequality gap this large by 2030 is not achievable through marginal policy adjustment; it requires the kind of transformational change (in taxation, labour markets, and social protection systems) that the SDG framework was explicitly designed to demand. Examiners should credit any two of the three elements provided they are correctly explained and explicitly linked back to the poverty/inequality tension in the question stem — the specific pair chosen is a candidate's choice, not a right/wrong answer.

Mark Allocation

Band	Range	Criteria
First Class	70–100%	Correctly names and explains two of the three elements (Universality / Integration / Transformation) with accurate SDG facts (17 goals, 169 targets, 2015 adoption), AND explicitly develops the link from each element to the poverty/inequality tension in the question stem, rather than describing the elements in the abstract.
Upper Second (2.1)	60–69%	Two elements correctly identified and explained, but the link to inequality/ poverty is asserted rather than developed with a specific mechanism (e.g. SDG 10, indivisibility of goals).
Second Class / Pass	40–59%	One element explained in reasonable depth, or two elements named with only superficial explanation; some factual imprecision about the SDG framework.
Fail	0–39%	SDGs described in general terms with no reference to the specific "elements" framework taught, or significant factual errors about the framework.

Question 3 total — 100% (50% of overall paper)

Question 4

50% of paper

QUESTION AS SET

"Food miles" is a term used to describe the distance that food is transported from the field or the producer to the plate or consumer. Discuss the role of food miles in food waste.

VERDICT

A real but limited and frequently overstated role. Food miles contribute to food waste mainly through post-harvest, transport-related spoilage in long or weak cold-chain supply chains — but the evidence does not support treating distance travelled as the dominant driver of either food waste or food-related emissions.

FULL MODEL ANSWER

1. Defining the concept

Food miles refers to the distance travelled, and the method of transport required, for food to move from where it is produced to where it is consumed. The term is widely used as an informal proxy for the environmental and food-security cost of a long supply chain, on the assumption that "further travelled" means "more wasted and more polluting."

2. Where food miles genuinely do contribute to food waste

There is a real, direct channel: perishable food travelling long distances through supply chains with weak infrastructure or a broken cold chain is more exposed to spoilage before it ever reaches a buyer. In India, for example, up to 40% of food rots on the way to market — a loss driven substantially by the distance and conditions of transport combined with inadequate storage and logistics. More broadly, of the roughly one-third of food produced for human consumption globally that is lost or wasted each year, losses occurring during transport, storage and distribution are a meaningful share of that total in developing-country supply chains in particular, where cold-chain infrastructure is limited. In this sense, food miles are not irrelevant to food waste — longer, less-controlled transport legs are a genuine opportunity for loss.

Sourcing note. *The India spoilage figure and the "roughly one-third of food produced is lost or wasted" estimate are drawn from the module's own food waste lecture content, which in turn cites Gustavsson et al. (2011), Global Food Losses and Food Waste: Extent, Causes and Prevention, FAO — the most widely cited study on global food loss and waste, independently verified as a genuine FAO publication.*

3. Why the role should not be overstated

The popular assumption that distance travelled is the main driver of food's environmental and waste footprint does not hold up well. Final delivery transport — the part of the journey most people picture when they hear "food miles" — accounts for only around 4% of food-related emissions; what is eaten and how much of it is wasted matters considerably more than where it travelled from. This is a critical distinction for the question: emissions and waste are related but separate outcomes, and food miles are a weak predictor of both. The concept has also been directly critiqued in the sustainability literature as an overly blunt indicator: MacGregor and Vorley's "Fair Miles" analysis argues that distance alone tells you little about a food product's true sustainability cost, since a food's total footprint depends heavily on production method, scale efficiency and storage requirements — a product grown locally in a heated greenhouse can carry a higher footprint than the same product shipped from a climate where it grows without artificial heating. Consumer research in the same vein finds that people who buy local often are not actually motivated by food miles as such, but by other concerns (freshness, supporting local producers), which further undermines food miles as a clean causal driver of behaviour, waste, or environmental outcomes.

> **Sourcing note.** *The "4% of food emissions" figure and the framing that eating/wasting behaviour matters more than distance are drawn from the module's own Food Miles, Losses and Wastage lecture slide. MacGregor & Vorley (2006), Fair Miles? The concept of "food miles" through a sustainable development lens, and the Sirieix et al. (2008) consumer-motivation study both appear in the module's own reading list and were independently verified as genuine published sources.*

4. Where the more significant sources of food waste actually sit

Along the food supply chain, larger and more consistent sources of waste sit outside the transport leg itself: retailer over-ordering and cosmetic/aesthetic rejection of produce, consumer over-purchasing and plate waste, and processing losses. Americans, for instance, are estimated to throw away up to 40% of the food they buy — a consumption-stage loss with essentially no relationship to how far that food travelled. This reinforces the verdict that food miles are one contributing factor among several, concentrated specifically in the transport-and-storage segment of the chain, rather than the primary explanation for food waste overall.

Conclusion: food miles play a genuine, identifiable role in food waste where long-distance transport intersects with weak cold-chain infrastructure, producing real post-harvest losses — particularly in developing-country supply chains. But the evidence does not support food miles as the dominant or even primary driver of food waste: emissions and waste are shaped far more by what is produced, how it is produced, and consumer and retail behaviour once it arrives, than by the distance it travelled to get there.

Mark Allocation

Band	Range	Criteria
First Class	70–100%	Defines food miles precisely; identifies a genuine transport-related mechanism into food waste (post-harvest spoilage in long/weak-cold-chain supply chains) with supporting evidence; critically evaluates the assumption that distance is the dominant driver, using evidence on emissions share and consumer/retail-stage waste; reaches a balanced, evidenced verdict rather than an unqualified "food miles cause waste" or "food miles are irrelevant" position.
Upper Second (2.1)	60–69%	Correct definition and at least one accurate mechanism linking food miles to waste, with some critical balance, but the critique of the "distance = footprint" assumption is thinner or under-evidenced.
Second Class / Pass	40–59%	Definition correct but discussion is largely descriptive; asserts a link between food miles and food waste without evidence or critical evaluation of the assumption.
Fail	0–39%	Definition of food miles absent or incorrect; no real engagement with the specific question of food miles' role *in food waste* as distinct from environmental impact generally.

Question 4 total — 100% (50% of overall paper)